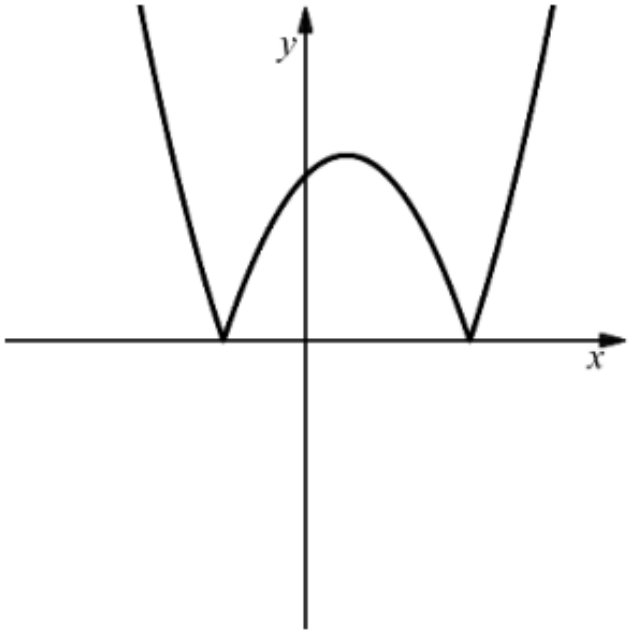
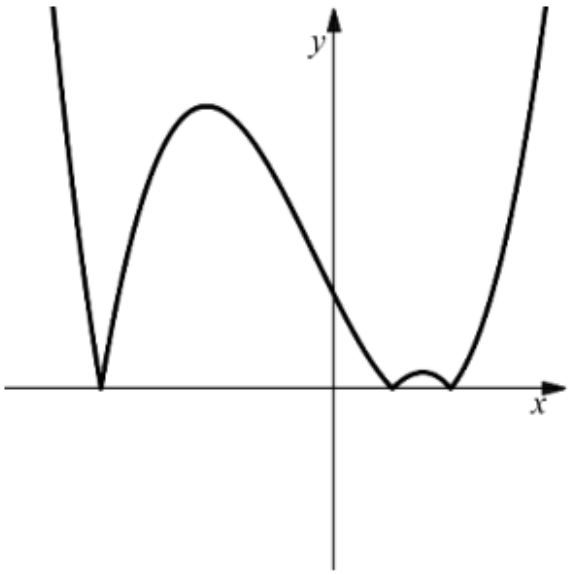
2.56 Graph Sketching: Modulus Functions

Worked Solutions

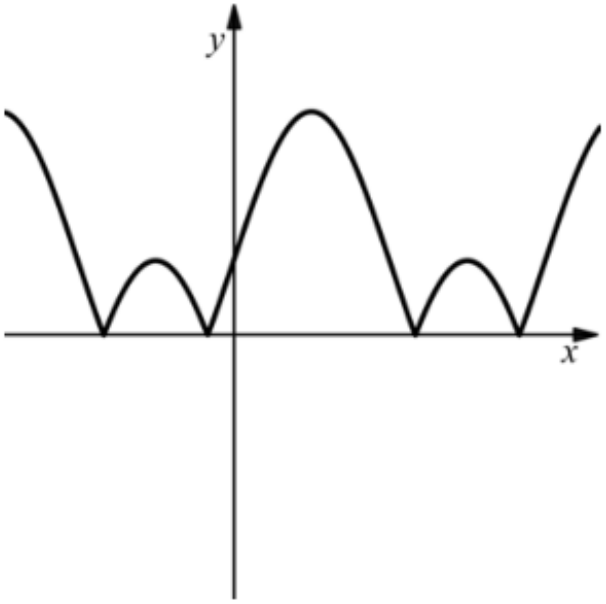
1 a)



1 b)

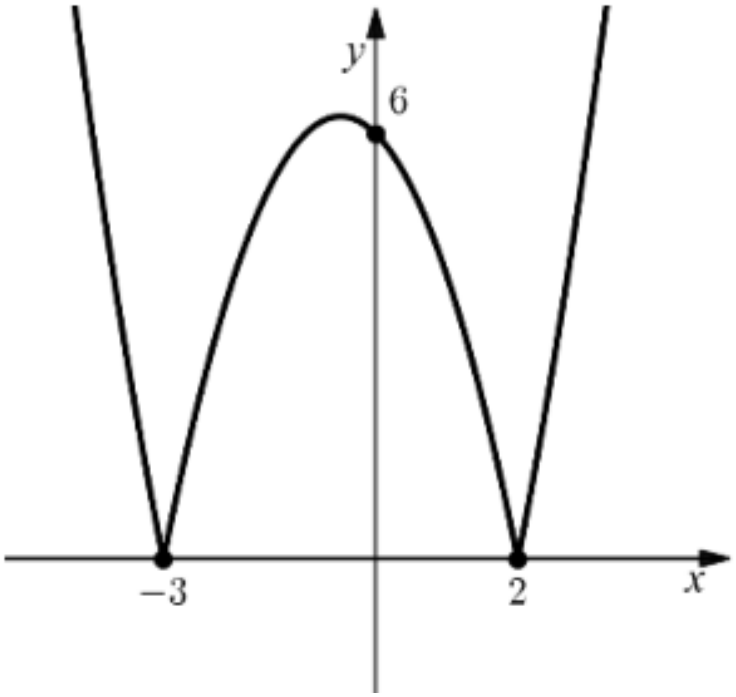


1 c)

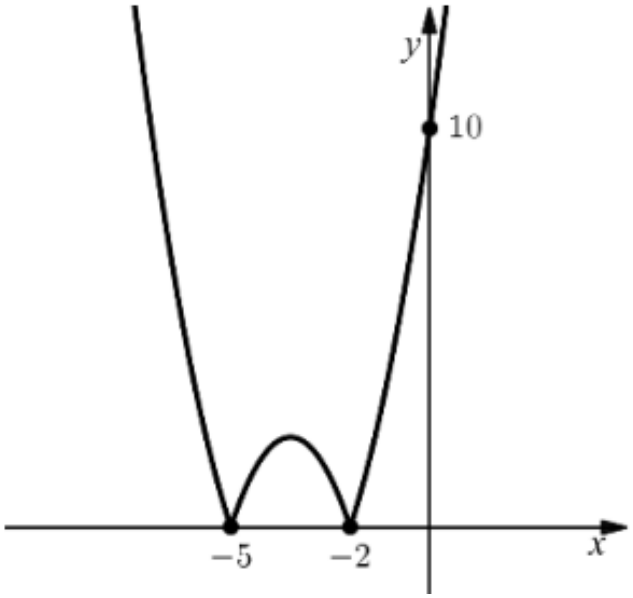


2 a)

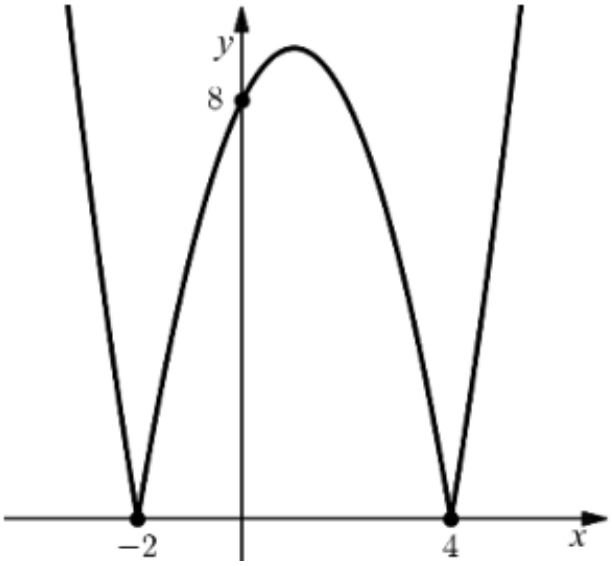
$x^2 + x - 6 = (x + 3)(x - 2)$.



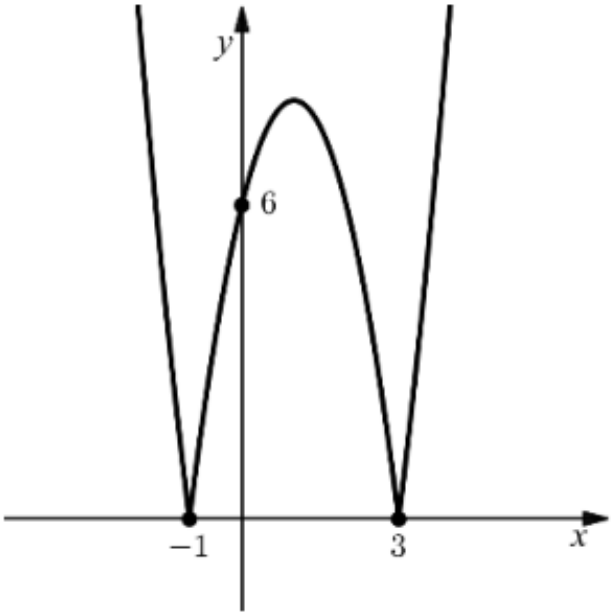
2 b)



2 c)



2 d)



3 a)

$k(x+1)(x-3)$, so $k = \pm 2$. Hence

$$\pm 2(x^2 - 2x - 3) = \pm(2x^2 - 4x - 6),$$

so $a = 2$, $b = -4$ and $c = -6$, or $a = -2$, $b = 4$ and $c = 6$.

3 b)

$kx(x-5)$, and $k\left(\frac{5}{2} - 5\right) = \frac{25}{4}$, so $k = -1$. Hence

$$\pm\left(x^2 - 5x\right) = \pm(x^2 - 5x),$$

so $a = 1$, $b = -5$ and $c = 0$, or $a = -1$, $b = 5$ and $c = 0$.

3 c)

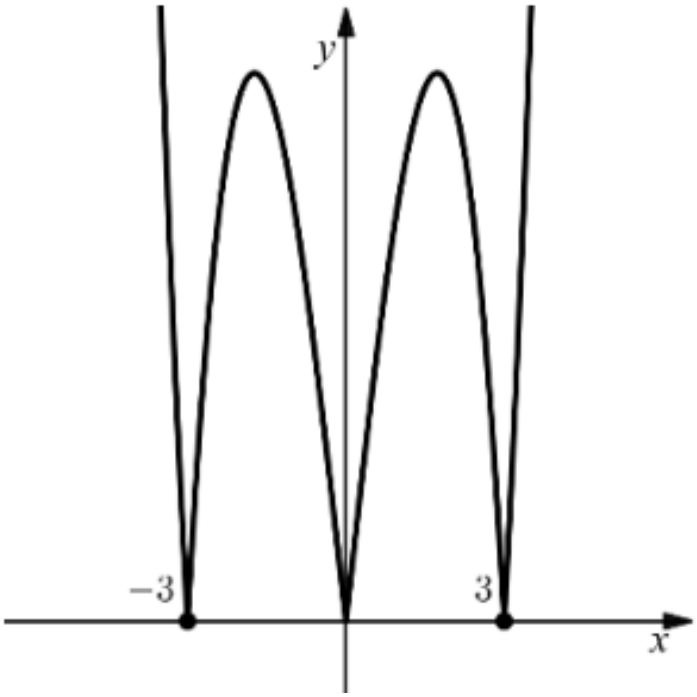
$k(x-2)^2$, so $k = 1$. Hence

$$\pm(x-2)^2 = \pm(x^2 - 4x + 4),$$

so $a = 1$, $b = -4$ and $c = 4$, or $a = -1$, $b = 4$ and $c = -4$.

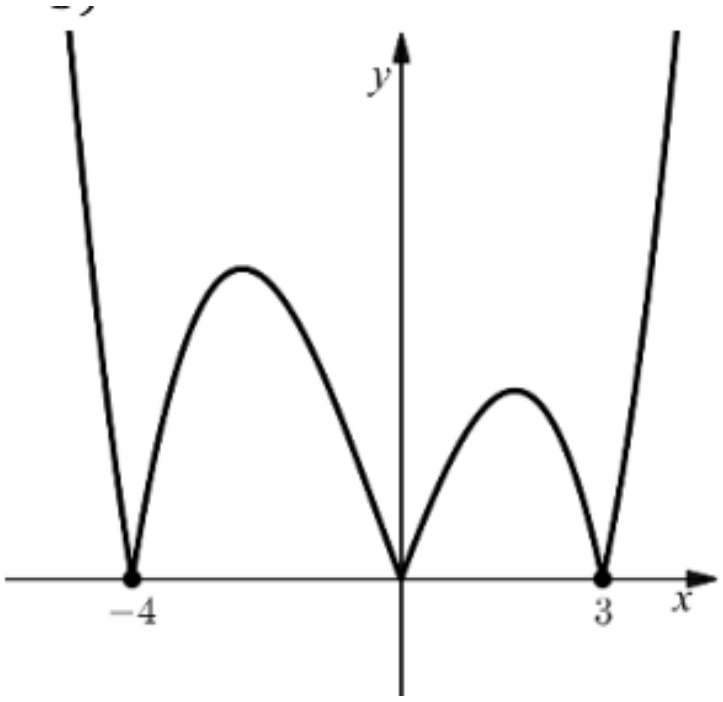
4 a)

$x^3 - 9x = x(x-3)(x+3)$.



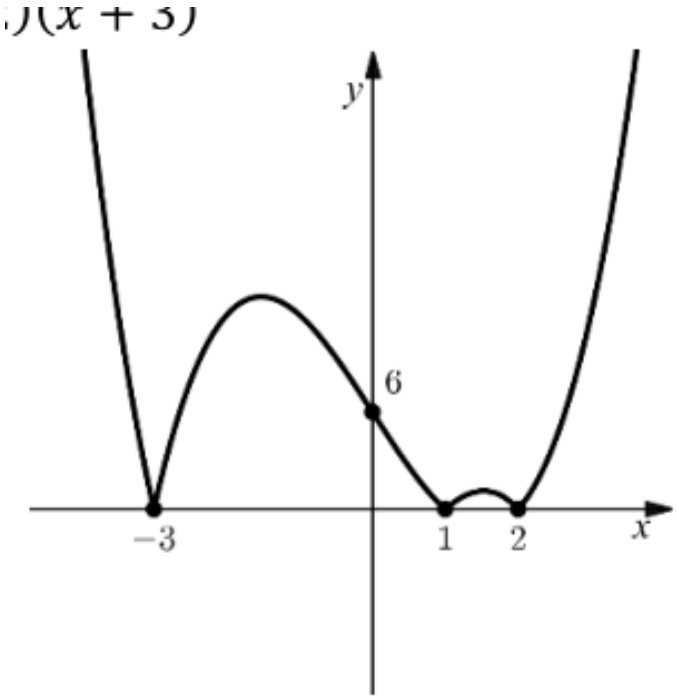
4 b)

$x^3 + x^2 - 12x = x(x+4)(x-3)$.



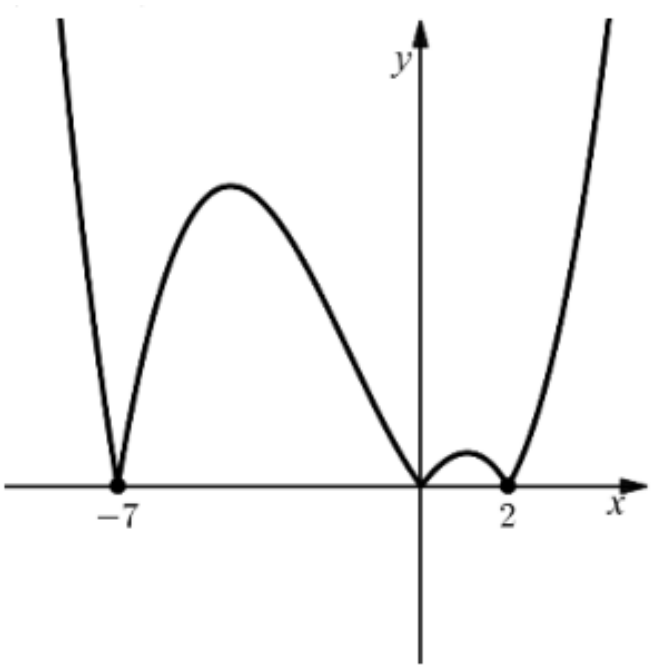
4 c)

$x^3 - 7x + 6 = (x-1)(x-2)(x+3)$.



4 d)

$14x - 5x^2 - x^3 = -x(x+7)(x-2)$.



5 a)

$k(x+3)(2x+1)(x-5)$, so $k = -1$. Thus

$$\pm(x+3)(2x+1)(x-5) = \pm(2x^3 - 3x^2 - 32x - 15),$$

so $a = 2$, $b = -3$, $c = -32$ and $d = -15$, or $a = -2$, $b = 3$, $c = 32$ and $d = 15$.

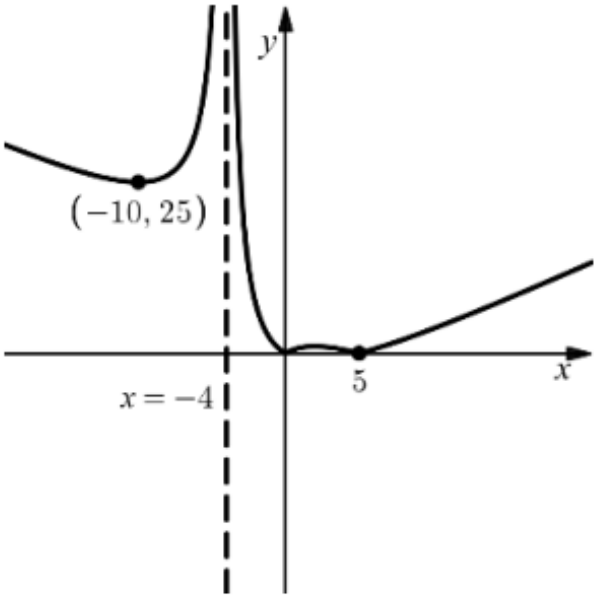
5 b)

$k(x+4)(3x+1)(x-2)$, so $k=-\frac{1}{2}$. Thus

$$\pm\frac{1}{2}(x+4)(3x+1)(x-2)=\pm\left(\frac{3}{2}x^3+\frac{7}{2}x^2-11x-4\right),$$

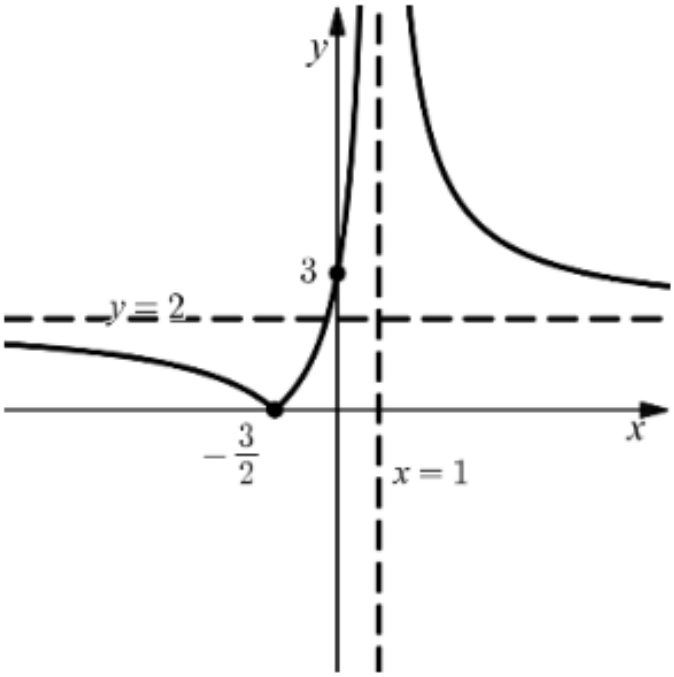
so $a=\frac{3}{2}$, $b=\frac{7}{2}$, $c=-11$ and $d=-4$, or $a=-\frac{3}{2}$, $b=-\frac{7}{2}$, $c=11$ and $d=4$.

6)



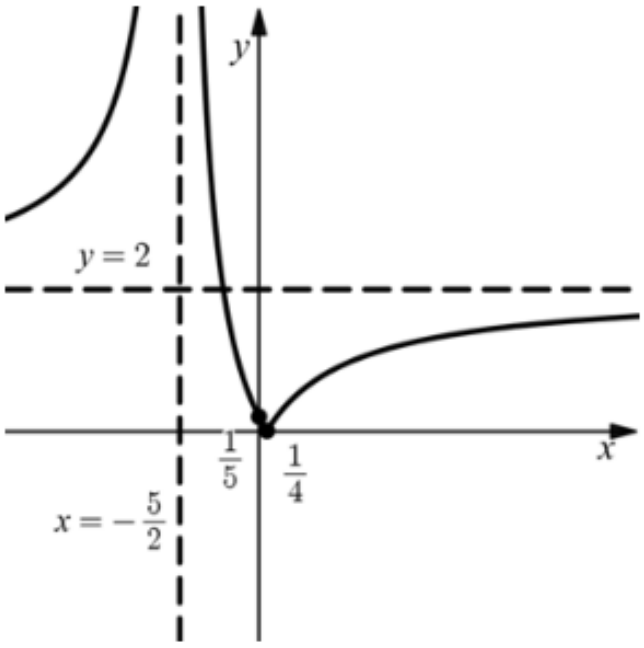
7 a)

Consider $y=\dfrac{2x+3}{x-1}$. When $x=0$, $y=-3$; when $y=0$, $x=-\frac{3}{2}$. The asymptotes are $x=1$ and $y=2$.

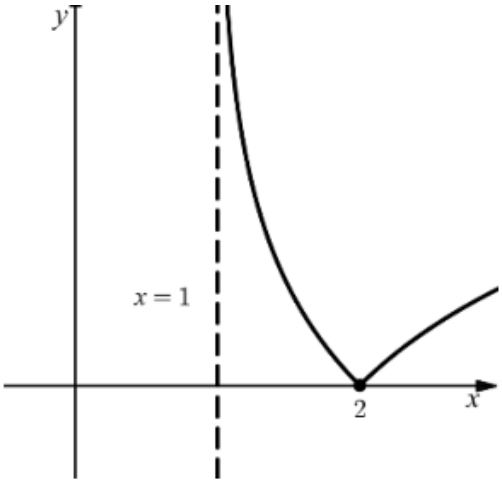


7 b)

Consider $y = \dfrac{1-4x}{2x+5}$. When $x = 0$, $y = \frac{1}{5}$; when $y = 0$, $x = \frac{1}{4}$. The asymptotes are $x = -\frac{5}{2}$ and $y = |-2| = 2$.

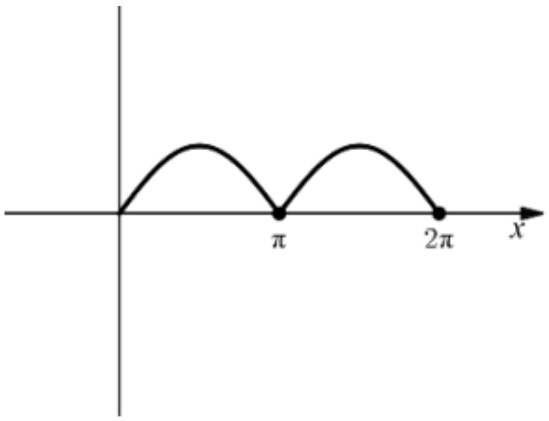


8 a)



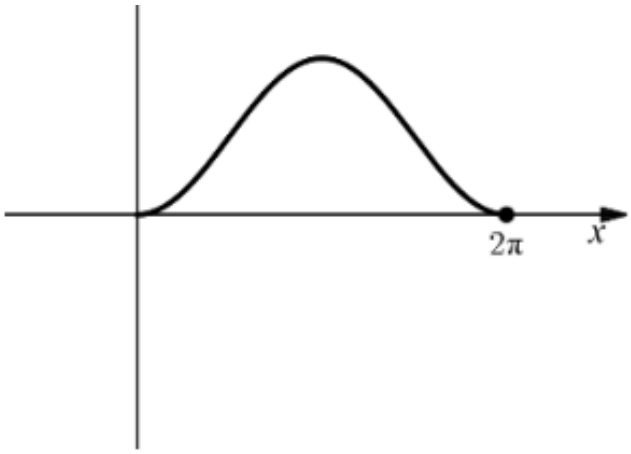


8 b)

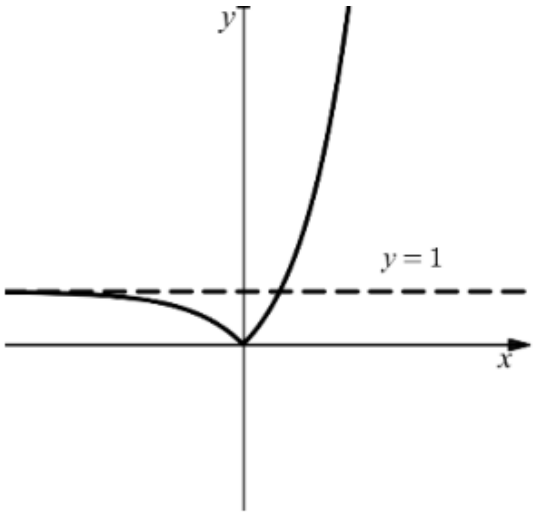




8 c)

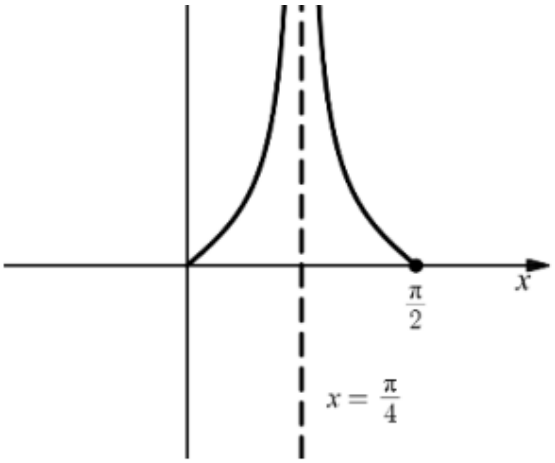


8 d)



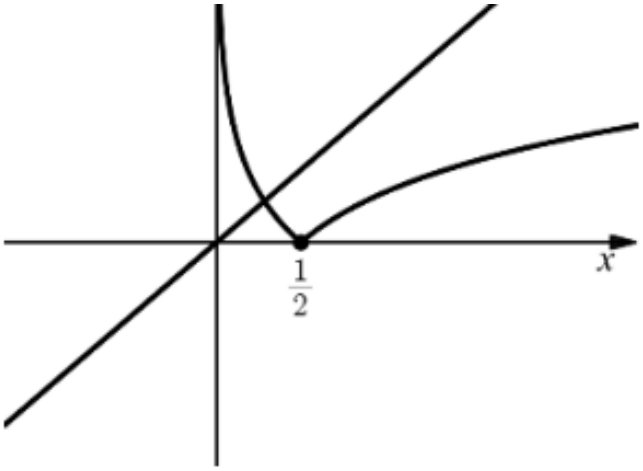


8 e)



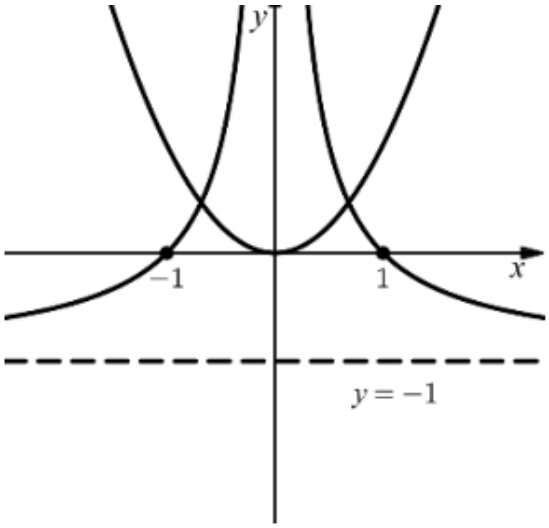


9 a)



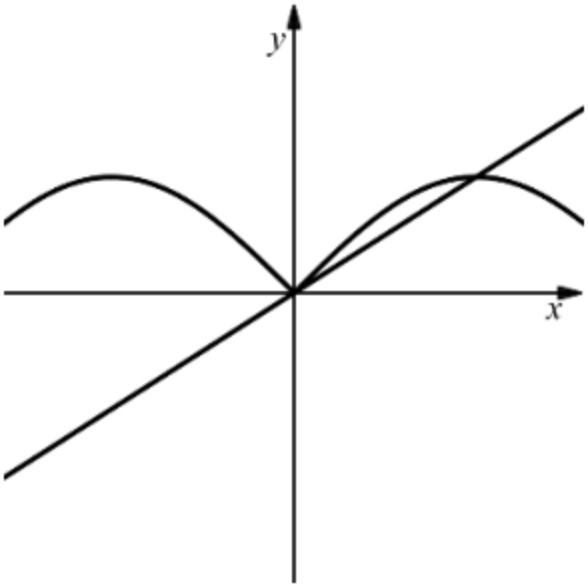
Therefore, there is 1 solution.

9 b)



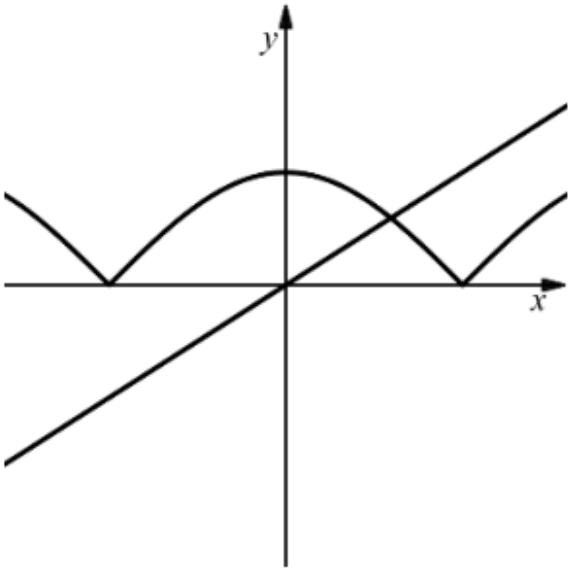
Therefore, there are 2 solutions.

9 c)



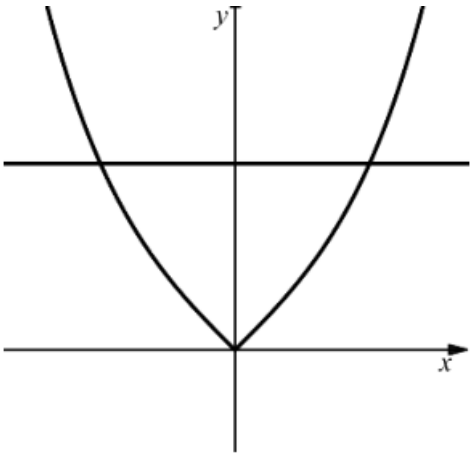
Therefore, there are 2 solutions.

9 d)



Therefore, there is 1 solution.

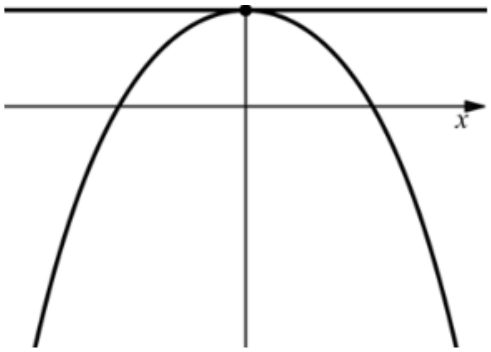
10 a)





Therefore, there are 2 solutions.

10 b)



$$\overline{^2+1}) = \ln(1+\sqrt{2})$$
$$\cdot\sqrt{2})$$
$$(1+\sqrt{2})$$

Therefore, there is 1 solution.

11 a)

$\sinh x = 1 \Rightarrow x = \ln(1+\sqrt{1^2+1}) = \ln(1+\sqrt{2})$. Then by symmetry, $-\ln(1+\sqrt{2})$. Hence $x = \ln(1+\sqrt{2})$ or $x = -\ln(1+\sqrt{2})$.

11 b)

$\cosh x - 2 = 1 \Rightarrow \cosh x = 3 \Rightarrow x = \ln(3+\sqrt{3^2-1}) = \ln(3+2\sqrt{2})$. Then by symmetry, $\ln(3-2\sqrt{2})$. Also, $\cosh x - 2 = -1 \Rightarrow \cosh x = 1 \Rightarrow x = \ln(1+\sqrt{1^2-1}) = 0$. Hence $x = \ln(3+2\sqrt{2})$, $x = \ln(3-2\sqrt{2})$ or $x = 0$.

12)

From the asymptote, $x = -\frac{1}{3}$. Then $c\left(-\frac{1}{3}\right) + d = 0$, so $d = \frac{c}{3}$. From the horizontal asymptote, $y = \frac{2}{3}$, so $\left|\frac{a}{c}\right| = \frac{2}{3}$. Take the smallest integers: $c = 3$, $d = 1$, $a = 2$. So far, $y = \left|\dfrac{2x+b}{3x+1}\right|$. The x-intercept is 2, so $2 \times 2 + b = 0$, giving $b = -4$. Therefore, $y = \left|\dfrac{2x-4}{3x+1}\right|$. Check the y-intercept: when $x = 0$, $y = \left|\dfrac{2\times 0-4}{3\times 0+1}\right| = 4$, as required. Hence $a = 2$, $b = -4$, $c = 3$ and $d = 1$.